

2018

HSC Trial Examinations

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Centre Number

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Student Number



Mathematics Extension 1 Trial

General Instructions

- Reading time – 5 minutes
- Working time 2 hours
- Write using blue or black pen
- NESA approved calculators may be used
- A formulae and data sheet is provided
- For questions 11 to 14, show relevant mathematical reasoning and/or calculations

Section I - 10 marks (Pages 3 – 6) • Attempt Questions 1 to 10 • Allow about 15 minutes for this section	Multiple Choice Questions	/10
	Question 11	/15
	Question 12	/15
	Question 13	/15
	Question 14	/15
Total		/70

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 40% of the HSC Course Assessment

Section I

10 marks

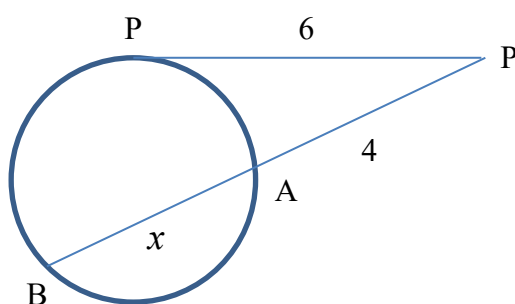
Attempt Questions 1 to 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 to 10 (Detach from paper)

1. Which of the following is a factor of $x^3 + 2x^2 - 7x + 4$?
 - (A) $x - 1$
 - (B) $x + 2$
 - (C) $x - 2$
 - (D) $x + 1$
2. Given that $N = 50 + 40e^{kt}$, which expression is equal to $\frac{dN}{dt}$?
 - (A) $k(50 - N)$
 - (B) $k(40 - N)$
 - (C) $k(N - 50)$
 - (D) $k(N - 40)$
3. If $A = \tan^{-1}(x)$, then the value of $\sin 2A$ is:
 - (A) $\frac{2x}{1-x^2}$
 - (B) $\frac{2x}{\sqrt{1-x^2}}$
 - (C) $\frac{2x}{\sqrt{1+x^2}}$
 - (D) $\frac{2x}{1+x^2}$
4. The domain and range of the function $f(x) = 2 \cos^{-1}\left(\frac{x}{3}\right)$ is given by:
 - (A) $0 \leq x \leq 3; -\pi \leq y \leq \pi.$
 - (B) $-3 \leq x \leq 3; 0 \leq y \leq \pi$
 - (C) $-3 \leq x \leq 3; 0 \leq y \leq 2\pi$
 - (D) $-\pi \leq x \leq \pi; 0 \leq y \leq 2$

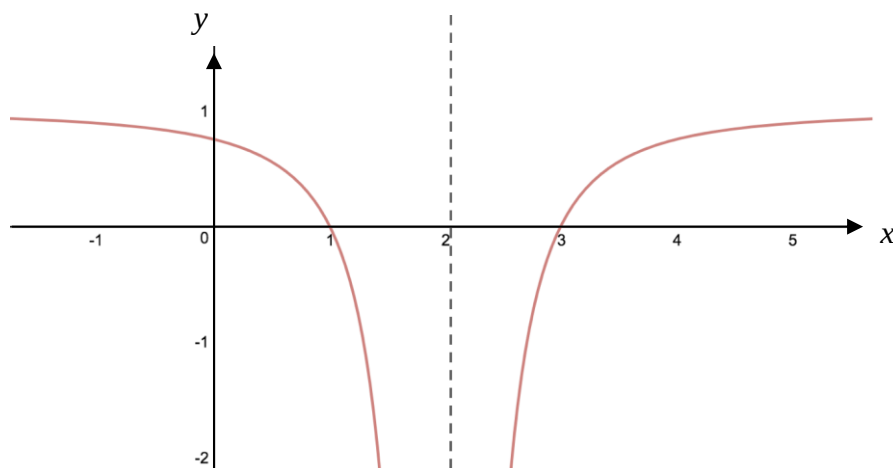
5. What is the x -intercept of the normal to the parabola $x^2 = 4ay$ at the point $(2ap, ap^2)$ on the parabola?
- (A) $ap(p^2 + 1)$
- (B) $ap(p^2 + 2)$
- (C) ap^2
- (D) $-ap^2$
6. A secant drawn from P to B and a tangent drawn from P to T meet the circle as shown below.



What is the value of the pronumeral x ?

- (A) $x = 9$
- (B) $x = \frac{3}{2}$
- (C) $x = 2$
- (D) $x = 5$
7. Find $\int \frac{e^{2x}}{1 + e^{2x}} dx$
- (A) $\tan^{-1}(e^x)$
- (B) $2 \tan^{-1}(e^{2x})$
- (C) $\frac{1}{2} \ln(1 + e^{2x})$
- (D) $2 \ln(1 + e^{2x})$

8.



Find the function $y = f(x)$ whose graph is given above:

(A) $f(x) = \frac{(x-1)(x-3)}{(x-2)^2}$

(B) $f(x) = \frac{2(x-1)(x-3)}{x-2}$

(C) $f(x) = \frac{2(x-1)(x-3)}{(x-2)^3}$

(E) $f(x) = \frac{2(x-1)(x-3)}{(x-2)^2}$

9.

The graph of $f(x) = 0.6 \cos^{-1}(x - 1)$, defines a curve that, when rotated about the y -axis will produce a solid that is to be the shape and size of a new biscuit. Which integral expression will give the volume of the biscuit?

(A) $\pi \int_0^{0.6} \left[\cos\left(\frac{3}{5}y\right) + 1 \right]^2 dy$

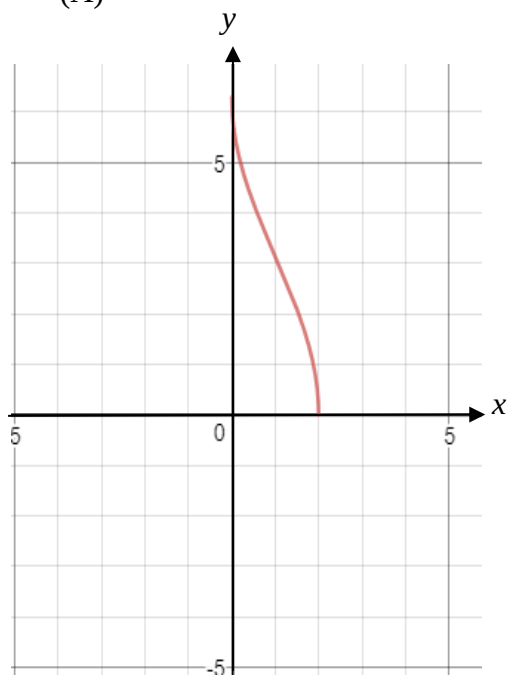
(B) $\pi \int_0^{0.6} \left[\cos\left(\frac{5}{3}y\right) + 1 \right]^2 dy$

(C) $\pi \int_0^{0.6\pi} \left[\cos\left(\frac{5}{3}y\right) + 1 \right]^2 dy$

(D) $\pi \int_0^{0.6\pi} \left[\cos\left(\frac{3}{5}y\right) + 1 \right]^2 dy$

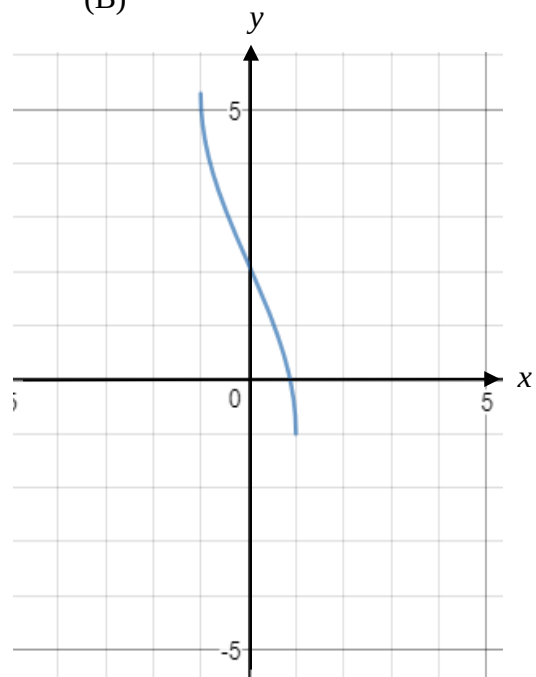
10. Which of the graphs match with the given equation?

(A)



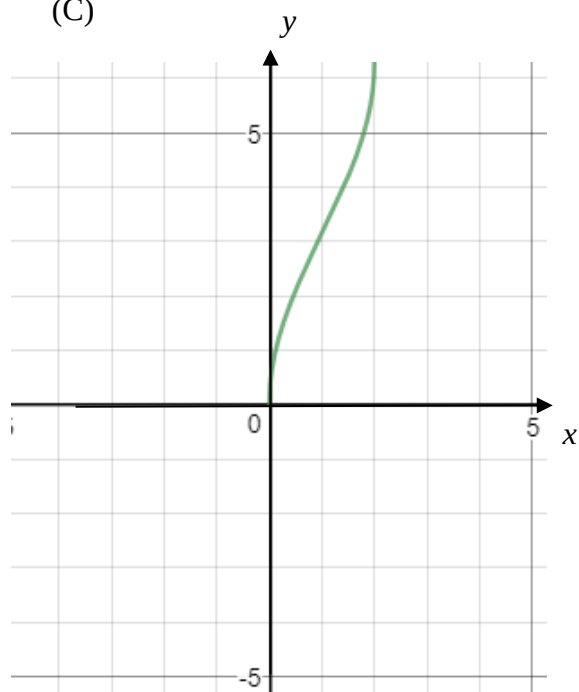
$$y = 2\cos^{-1}(x + 1)$$

(B)



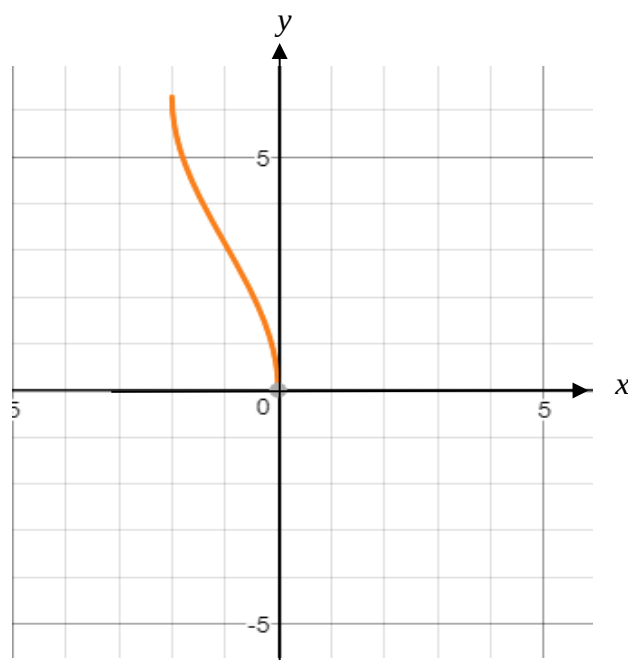
$$y = 2\cos^{-1}(x - 1)$$

(C)



$$y = 2\cos^{-1}(1 - x)$$

(D)



$$y = 2\cos^{-1}x - 1$$

Section II

90 marks

Attempt Questions 11 to 14

Allow about 1 hours and 45 minutes for this section

Answer each question in a **separate writing booklet**. Extra writing booklets are available.

In Questions 11 to 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\int_0^{\frac{\pi}{4}} \cos^2 2x \, dx$ 3

(b) The acute angle between $5x - y - 3 = 0$ and $y = mx + 4$ is 45° .
Find two possible values of m . 2

(c) Solve $\frac{2x}{x-2} \geq 1$ 3

(d) Express $\cos x + \sin x$ in the form $R \cos(x + \alpha)$, where α is in radians. 2

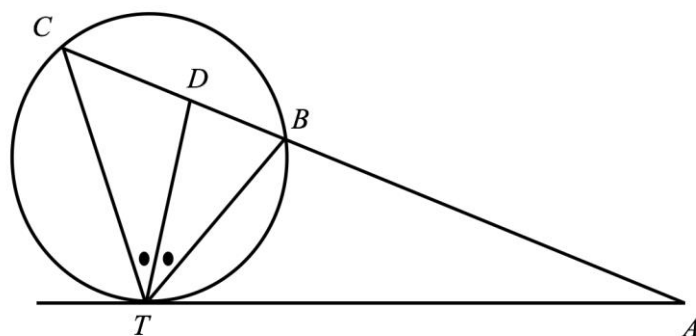
(e) Evaluate $\int_1^2 \frac{4x}{\sqrt{x^2-1}} \, dx$ using the substitution $u = x^2 - 1$. 3

(f) Let α, β, γ be roots of $P(x) = x^3 - 2x^2 - x + 2$. Evaluate $\alpha^2 + \beta^2 + \gamma^2$
given that $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ 2

Question 12 (15 marks) Use a SEPARATE writing booklet

(a)

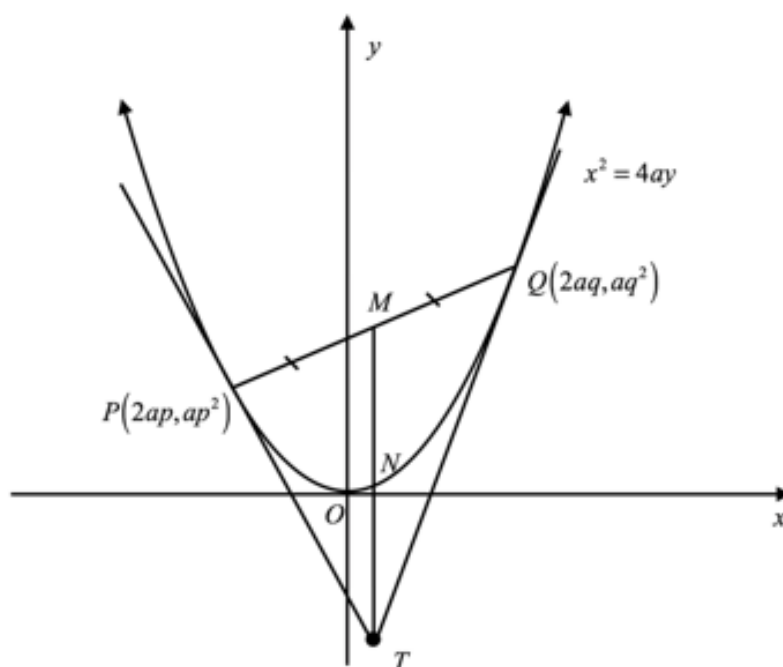
3



TA is a tangent to a circle. Line $ABDC$ intersects the circle at B and C . Line TD bisects angle BTC .

Prove $AT = AD$

(b)



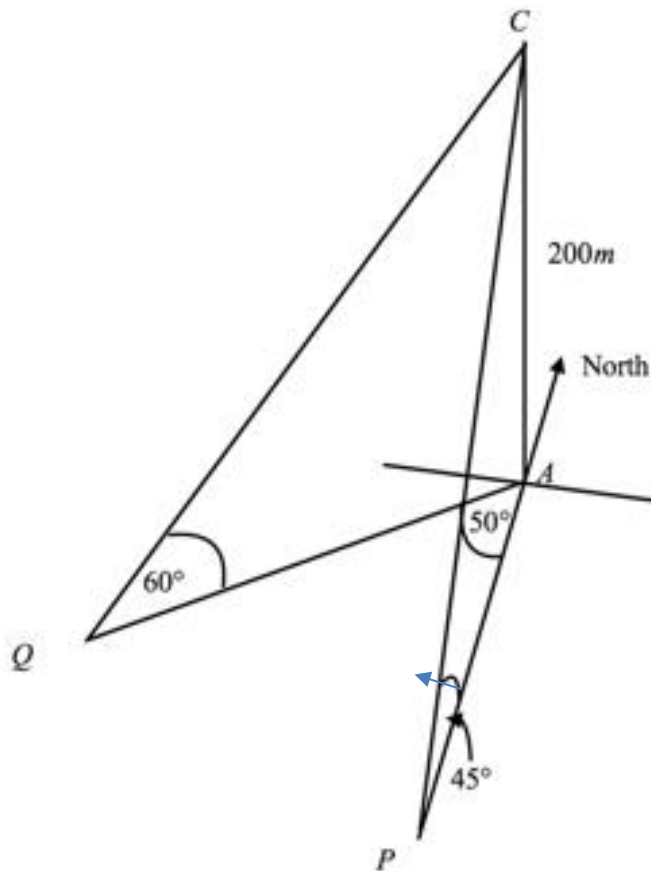
In the diagram, $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

- i) Show that the tangent at P has equation $y = px - ap^2$. 2
- ii) The tangent at P and Q meet at T . Assuming that the tangent at Q is $y = qx - aq^2$, show that T is the point $(a(p + q), apq)$. 2
- iii) M is the midpoint of the chord PQ . Show that MT is parallel to the axis of symmetry of the parabola. 2

Question 12 continues on page 8

Question 12 continued

- (c) From the top, C , of a vertical cliff, 200 m high, two ships P and Q are observed at sea level. A is the foot of the cliff at sea level. P is due south of A and the angle of elevation of C from P is 45° . Q is $S 50^\circ W$ of A and the angle of elevation of C from Q is 60° . 3



Find the distance PQ (to the nearest metre).

- (d) i) By considering the graph of $y = e^x$, show that the equation $e^x + x + 1 = 0$ has only one real root and that this root is negative. 2
- ii) Taking $x = -1.5$ as a first approximation to this root, use one application of Newton's method to find a better approximation. (Give your answer correct to three decimal places) 2

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) Use the t -formula to solve for x (to the nearest minute).

$$7\sin x - 4\cos x = 4, \quad \text{for } 0^\circ \leq x \leq 360^\circ \quad 3$$

- (b) Prove by mathematical induction that

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1} \quad 3$$

For all integers $n \geq 1$.

- (c) Consider the function

$$f(x) = 2 \sin^{-1} \sqrt{x} - \sin^{-1}(2x - 1)$$

- i) Find the domain of $f(x)$ 1

- ii) Show that $f'(x) = 0$ 3

- iii) Sketch the graph of $y = f(x)$ 1

- (d) The rise and fall of a tide can be modelled as simple harmonic motion. A cruise ship needs 11 metres of water to pass down a channel safely. At low tide the channel is 8 metres deep and at high tide it is 12 metres deep. Low tide is at 10 am and high tide is 4pm.

- (i) Show the water depth, y metres, in the channel, is given by 2

$$y = 10 - 2 \cos\left(\frac{\pi t}{6}\right)$$

- (ii) Find the earliest time period after 10am (i.e. between which two times) that the cruise ship can safely proceed through the channel. 2

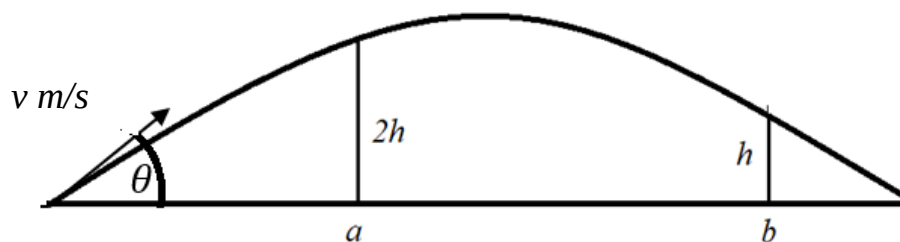
Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) On a tropical island that is being set up as a nature reserve, there are initially 100 nesting pairs of seagulls. In the 1st year, the number of pairs increases by 8. One theory suggested that the number N of nesting pair after t years would satisfy the differential equation $\frac{dN}{dt} = \frac{1}{k}N(500 - N)$.
- i) Show that the value of k is 5000. 1
- ii) Given that $\frac{10a}{x(a-x)} = \frac{10}{x} + \frac{10}{a-x}$ show that the differential equation 3
can be expressed in the form $\frac{dt}{dN} = \frac{10}{N} + \frac{10}{500-N}$ and find the
equation for t in term of N .
- iii) Predict, using your answer in part (ii), after how many years (to the 1
nearest year) the number of pairs of nesting seagulls on the island
will be 300.
- (b) i) The acceleration of a particle moving in the x –axis is numerically
equal to $a = v^2(1 - v) \text{ cm s}^{-2}$.
Initially $x = 0 \text{ cm}$, $v = 0.5 \text{ cm s}^{-1}$.
- Show that $v = \frac{e^x}{1+e^x}$ (Hint: $\frac{1}{v(1-v)} = \frac{1}{v} + \frac{1}{1-v}$) 2
- ii) Show that $t = x + 1 - e^{-x}$. 2
- iii) Hence determine the range of values of x and v . 2

Question 14 continues on page 10

(c) **Question 14 continued**

A particle is projected at an angle θ to the horizontal with a velocity of $V \text{ m s}^{-1}$ so as to just clear two walls. The walls of height $2h$ metres and h metres are at distances a metres and b metres respectively from the point of projection.



- i) Given that $x = V \cos \theta t$ and $y = V \sin \theta t - \frac{1}{2} g t^2$, where x is the horizontal displacement and y is the vertical displacement in t seconds, show that $y = x \tan \theta - \frac{g x^2}{2 V^2} (1 + \tan^2 \theta)$ 1
- ii) Hence show that $\tan \theta = \frac{h(2b^2 - a^2)}{ab(b - a)}$ 3

END OF EXAMINATION

Q1 A Q2 C Q3 D Q4 (C) Q5 B Q6 D Q7 C Q8 A Q9 C Q10 C

1.	$P(.) = 1^3 + 2 \times 1^2 - 7 \times 1 + 4$ $= 1 + 2 - 7 + 4 = 0$ $\therefore (x-1)$ is a factor $\therefore$ (A)	5.	$x + py = 2ap + ap^3$ $\therefore y = 0, x = 2ap + ap^3$ B
2.	$N = 50 + 40 \cdot e^{kt}$ $\frac{dN}{dt} = 40k \cdot e^{kt}$ $= k \cdot (40e^{kt})$ $a_a \quad N = 50 + 40e^{kt}$ $40e^{kt} = (N-50)$ $\therefore \frac{dN}{dt} = k(N-50)$ $\therefore$ (C)	6.	$r\rho^2 = rA \cdot \rho B$ $6^2 = 4(4+x)$ $36 = 16 + 4x$ $\therefore 4x = 20$ $\therefore x = 5 \quad \therefore$ (D)
3.	$\sin 2A = 2 \sin A \cos A = \frac{2}{\sqrt{x^2+1}} \cdot \frac{2}{\sqrt{x^2+1}} = \frac{2x}{1+x^2}$ (D)	7.	C
4.	(C)	8.	A
		9.	$y = \frac{\pi}{4} - \tan^{-1} \frac{x}{3}$ R: $-\frac{\pi}{4} < y < \frac{3\pi}{4}$ (C)
		10.	$y = \frac{3}{5} \cos^{-1}(x-1)$ $\therefore x = \cos\left(\frac{5}{3}y\right) + 1$ $0 \leq \frac{5}{3}y = \cos^{-1}(x-1) \leq \pi$ $\therefore 0 \leq y \leq \frac{3}{5}\pi$ C

Q11a

$$\int_0^{\frac{\pi}{4}} \cos^2 2x \, dx$$

$$\text{ad } \cos 2x = 2\cos^2 x - 1$$

$$\Rightarrow \cos 4x = 2\cos^2(2x) - 1$$

$$\therefore 2\cos^2(2x) = \cos 4x + 1$$

$$\cos^2 2x = \frac{1}{2}(\cos 4x + 1)$$

$$\therefore \int_0^{\frac{\pi}{4}} \cos^2 2x \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} (\cos 4x + 1) \, dx \quad [1 \text{ mark}]$$

$$= \frac{1}{2} \left[\frac{1}{4} \sin 4x + x \right]_0^{\frac{\pi}{4}} \quad [1 \text{ mark}]$$

$$= \frac{1}{2} \left[\left(\frac{1}{4} \sin \pi + \frac{\pi}{4} \right) - \left(\frac{1}{4} \sin 0 + 0 \right) \right] \quad \underline{\text{note}} \sin \pi = 0$$

$$= \frac{\pi}{8} \quad [1 \text{ mark}]$$

11 b.

$$5x - y - 3 = 0$$

$$y = m_1x + 4$$

$$y = 5x - 3$$

$$\therefore m_1 = 5$$

$$\therefore m_2 = m$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{5 - m}{1 + 5m} \right|$$

$$\therefore 1 = \left| \frac{5 - m}{1 + 5m} \right|$$

[1 mark]

$$1 = \frac{5 - m}{1 + 5m}$$

$$-1 = \frac{5 - m}{1 + 5m}$$

$$1 + 5m = 5 - m$$

$$-1 - 5m = 5 - m$$

$$\therefore 6m = 4$$

$$\therefore -6 = 4m$$

$$\therefore m = \frac{2}{3}$$

$$\therefore m = -\frac{3}{2}$$

$$\therefore m = \frac{2}{3}, -\frac{3}{2}$$

[1 mark for both values of m]

11c

Solve $\frac{2x}{x-2} \geq 1$

note $x \neq 2$

$$(x-2)^x \cdot \frac{2x}{(x-2)} \geq 1(x-2)^2 \quad [1 \text{ mark}]$$

$$(x-2)2x \geq (x-2)^2$$

$$0 \geq (x-2)^2 - (x-2)2x$$

$$\therefore 0 \geq (x-2)[(x-2)-2x]$$

$$0 \geq (x-2)(-x-2)$$

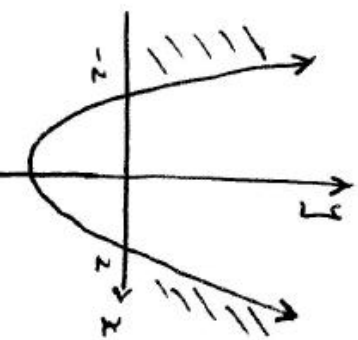
$$(0 \geq (x-2) \cdot -(x+2)) \quad x-1$$

$$(x-2)(x+2) \geq 0$$

[1 mark for correct rearranging & simplifying]

$$\therefore x \leq -2, x \geq 2$$

noting $x \neq 2$. } [1 mark]



11d $\cos x + \sin x = R \cos(x + \alpha)$

Note: $\cos(x + \alpha) = \cos x \cos \alpha - \sin x \sin \alpha$

$$\therefore \cos x + \sin x = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\therefore R \cos \alpha = 1$$

$$R \sin \alpha = -1$$

$$\therefore \frac{R \sin \alpha}{R \cos \alpha} = \frac{-1}{1} \Rightarrow \tan \alpha = -1$$

$$\therefore \alpha = \tan^{-1}(-1) = \frac{-\pi}{4} \quad [1 \text{ mark}]$$

$$R = \sqrt{(1)^2 + (-1)^2} = \sqrt{2} \quad (\text{By Pythagoras}) \quad [1 \text{ mark for } R]$$

$$\therefore \cos x + \sin x = \sqrt{2} \cos\left(x - \frac{\pi}{4}\right)$$

11e

$$\int_1^2 \frac{4x}{\sqrt{x^2-1}} dx \quad \text{using } u = x^2-1$$

$$\text{if } u = x^2-1$$

$$\frac{du}{dx} = 2x$$

$$\therefore du = 2x dx$$

$$\left. \begin{array}{l} \text{terminals} \\ u = x^2-1 = 3 \\ u = 1^2-1 = 0 \end{array} \right\} \quad \left[\begin{array}{l} 1 \text{ mark, terminals} \\ \text{and } \frac{du}{dx} \end{array} \right]$$

$$\therefore \int_1^2 \frac{4x}{\sqrt{x^2-1}} dx$$

$$= 2 \int_1^2 \frac{2x}{\sqrt{x^2-1}} dx$$

$$= 2 \int_0^3 \frac{1}{\sqrt{u}} du$$

$$= 2 \int_0^3 u^{-1/2} du$$

$$= 4 [u^{1/2}]_0^3 \quad [1 \text{ mark}]$$

$$= 4(\sqrt{3}-0)$$

$$= 4\sqrt{3} \quad [1 \text{ mark}]$$

11 F

$$P(x) = x^3 - 2x^2 - x + 2$$

$$\alpha + \beta + \gamma = \frac{-b}{a} = 2$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -1$$

} [1 mark]

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 2^2 - 2(-1)$$

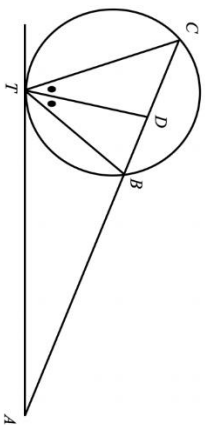
$$= 4 + 2$$

$$= 6$$

[1 mark]

(a)

3



TA is a tangent to a circle. Line $ABDC$ intersects the circle at B and C . Line TD bisects angle BTC .

Prove $AT = AD$

3 marks for conclusion with reasoning and working

2 marks for significant progress to solution, finding multiple other angles with reasoning, or equivalent

let $\angle ATB = \alpha$

let $\angle BTD = \beta$

$\angle ACT = \alpha$ (angle between a tangent and a chord is equal to the angle on the circumference subtended by the chord in the alternate segment)

$\therefore \angle CAT = 180 - (\alpha + 2\beta) - \alpha$ (angle sum of a triangle)

similarly,

$\angle ADT = 180 - \angle CAT - (\alpha + \beta)$

$$= 180 - 180 + (\alpha + 2\beta) + \alpha - (\alpha + \beta)$$

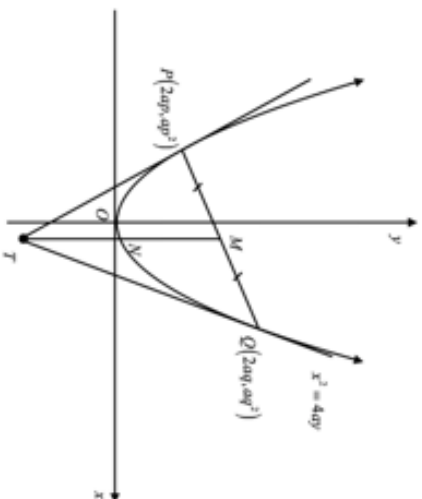
$$= \alpha + \beta$$

$$\therefore \angle DTA = \alpha + \beta = \angle ADT$$

$\therefore \triangle DTA$ is isosceles as 2 angles are equal

$\therefore AT = AD$ (angles opposite equal sides are equal in isosceles triangle)

(b)



(i)

2 marks for correct working towards solution

1 mark for differentiating the parabola or equivalent

$$x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{x}{2a}$$

at point $P(2ap, ap^2)$ $y' = p$

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = p(x - 2ap)$$

$$y = px - 2ap^2 + ap^2$$

$$y = px - ap^2$$

as required

(ii)

2 marks for correct and complete working

1 mark for solving simultaneously to find x or y

we have:

$$y = px - ap^2$$

$$y = qx - aq^2$$

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p - q) = ap^2 - aq^2$$

$$x(p - q) = a(p - q)(p + q)$$

$$x = a(p + q)$$

$$y = p(a(p + q)) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$$\therefore (a(p + q), apq)$$

(iii)

2 marks for correct answer with sound reasoning

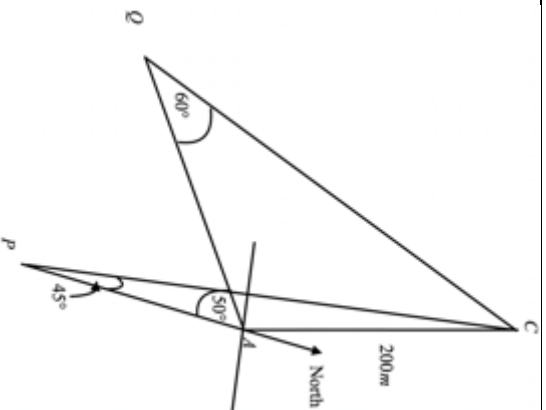
1 mark for finding the x value of M or equivalent

x value given by

$$x = \frac{2ap + 2aq}{2}$$

$$= a(p + q)$$

As M and T have the same x value, the line MT is vertical, as is the axis of the parabola, hence parallel.



3 marks for correct answer with correct and complete working

2 marks finding expressions for PA and QA and an attempt to use them

$$\tan 45^\circ = \frac{200}{PA}$$

$$\tan 60^\circ = \frac{200}{QA}$$

$$PA = \frac{200}{\tan 45^\circ}$$

$$QA = \frac{200}{\tan 60^\circ}$$

$$PA = 200$$

$$QA = \frac{200}{\sqrt{3}}$$

by the cosine rule,

$$QP^2 = PA^2 + QA^2 - 2 \cdot PA \cdot QA \cdot \cos 50^\circ$$

$$QP^2 = 23644.24537...$$

$$QP = 153.766854...$$

therefore QP has length 154 metres

(d) i)

2 marks for graph with both lines and clearly one intercept

1 mark for either line drawn correctly on scaled axes, or equivalent

sketch both $y = e^x$ and $y = -x - 1$

We can see there is only one intercept, hence only one solution

(ii)

2 marks for correct answer with correct working

1 mark for correct substitution or correct derivative, but not both

$$y = e^x + x + 1$$

$$y' = e^x + 1$$

by Newton's Method

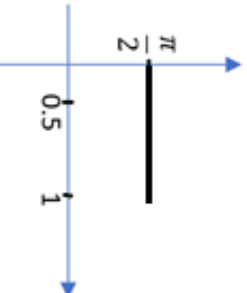
$$x_2 = x_1 - \frac{f(x)}{f'(x)}$$

$$x_2 = -1.5 - \frac{e^{-1.5} - 1.5 + 1}{e^{-1.5} + 1}$$

$$x_2 = -1.273638286...$$

$$x_2 = -1.274(3dp)$$

(a) $\frac{14t}{1+t^2} - \frac{4(1-t^2)}{1+t^2} = 4$ $14t - 4 + 4t^2 = 4 + 4t^2$ $14t = 8 \quad \rightarrow \quad t = \frac{4}{7}$ $\tan\left(\frac{\theta}{2}\right) = \frac{4}{7} \quad \rightarrow \quad \frac{\theta}{2} =$ Test for $\theta = 180^\circ$: LHS = $7 \times 0 + 4 \times 1 = 4 = RHS$ $\therefore \theta = 180^\circ, \dots$	1 mark-using correct formula 1 mark – correct answer of θ. 1 mark – testing for solution 180°	
(b) Prove that it is true for $n = 1$ $LHS = \frac{1}{1 \times 2} = \frac{1}{2}$ $RHS = 1 - \frac{1}{2} = \frac{1}{2}$ LHS = RHS $\therefore$ it is true for $n = 1$ Assume that it is true for $n = k$ i.e. $\frac{1}{1 \times 2} + \frac{1}{1 \times 3} + \frac{1}{1 \times 4} + \dots + \frac{1}{k(k+1)} = 1 - \frac{1}{k+1}$ RTP that it is true for $n = k + 1$ i.e. proving $\frac{1}{1 \times 2} + \frac{1}{1 \times 3} + \frac{1}{1 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+2)(k+1)} = 1 - \frac{1}{k+2}$ (3) By assumption $\frac{1}{1 \times 2} + \frac{1}{1 \times 3} + \frac{1}{1 \times 4} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+2)(k+1)} = 1 - \frac{1}{k+1} + \frac{1}{(k+2)(k+1)}$ $RHS = 1 - \frac{(k+2)-1}{(k+1)(k+2)}$ $= 1 - \frac{(k+1)}{(k+1)(k+2)}$ $= 1 - \frac{1}{k+2} = RHS \text{ of (3)}$ $\therefore$ If it is true for $n = k$ then it is true for $n = k + 1$ Hence the statement is true for $n = 1$ and it is true for $n = 2$ and so on. By Mathematical Induction it is true for all integer $n > 0$.	1 mark – proving $n = 1$ 2 – correct steps to achieve LHS = RHS 1 – working toward answer. -1 – if no conclusion or conclusion does not make sense.	

(C) (i) Domain: $0 \leq x \leq 1$ and $-1 \leq 2x - 1 \leq 1$ $0 \leq x \leq 1$ and $0 \leq x \leq 1$ $\therefore 0 \leq x \leq 1$ (ii) $\frac{d}{dx} f(x) = \frac{2}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}} - \frac{1}{\sqrt{1-(2x-1)^2}} \times 2$ $= \frac{1}{\sqrt{x}\sqrt{1-x}} - \frac{1}{\sqrt{4x(1-x)}}$ $= \frac{1}{\sqrt{x}\sqrt{1-x}} - \frac{1}{\sqrt{x}\sqrt{1-x}} = 0$ (iii) When $x = 0$, $f(0) = 2 \sin^{-1} \sqrt{0} - \sin^{-1}(-1)$ $= \frac{\pi}{2}$	1 marks – correct domain with working 2 marks – correct differentiation of the two functions. 1 mark – simplify to achieve correct answer. 1 mark – correct graph with labelling as shown.	
		

(D) (i)  From the diagram: centre of motion = $\frac{12+8}{2} = 10$ Amplitude is 2 Period: 10am $\rightarrow$ 4pm is 6 hours ie $\frac{1}{2}$ period. $\therefore 2 \times 6 = \frac{2\pi}{n} \rightarrow n = \frac{\pi}{6}$ The equation of motion is $y = 10 - 2 \cos\left(\frac{\pi t}{6}\right)$ (ii) When $y = 11 \rightarrow 11 = 10 - 2 \cos\left(\frac{\pi t}{6}\right)$ $\cos\left(\frac{\pi t}{6}\right) = -\frac{1}{2}$ $\frac{\pi t}{6} = \frac{2\pi}{3}, \frac{4\pi}{3} \rightarrow t = 4 \text{ and } 8 \text{ hours}$ $\therefore$ it is between 2 pm and 6pm.	2 marks – correct answer with diagram and working shown. 1mark – correct amplitude & period. 1 – correct equation of motion 1 – correct answers	
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Q19 (a)

When $n = 100$, $\frac{dN}{dt} = 8$

$$\rightarrow 8 = \frac{1}{k} \times 100(500 - 100)$$

$$8 = \frac{40000}{k} \rightarrow k = 5000$$

$$(ii) \quad \frac{10k}{N(k-n)} = \frac{10}{N} + \frac{10}{k-n}$$

When $k = 500$ and $k = N$

$$\frac{5000}{N(500-N)} = \frac{10}{N} + \frac{10}{500-N}$$

$$(ie) \quad \frac{dt}{dN} = \frac{10}{N} + \frac{10}{500-N}$$

Integrating both sides w.r.t N ,

$$\int \frac{dt}{dN} dN = \int \left(\frac{10}{N} + \frac{10}{500-N} \right) dN$$

$$t = 10 \ln \left(\frac{N}{500-N} \right) + C$$

When $t = 0$, $N = 100$

$$\Rightarrow 0 = 10 \ln \left(\frac{100}{400} \right) + C$$

$$C = 10 \ln(4)$$

$$\Rightarrow t = 10 \ln \left(\frac{4N}{500-N} \right)$$

$$(iii) \quad \text{When } N = 300,$$

$$t = 10 \ln \left(\frac{4 \times 300}{500-300} \right)$$
$$= 10 \ln(6) \text{ years}$$
$$\approx$$

(b)

$$(i) \quad \frac{dv}{dx} = v \frac{dv}{dx} = v^2 (1-v)$$

$$\Rightarrow \int \frac{1}{v(1-v)} dv = \int dx$$

$$\int \frac{1}{1-v} + \frac{1}{v} dv = \int dx$$

$$\ln\left(\frac{v}{1-v}\right) = x + C$$

$$x \rightarrow 0, v \rightarrow 0.8$$

$$\therefore C = 0$$

$$\Rightarrow \ln\left(\frac{v}{1-v}\right) = x$$

$$\frac{v}{1-v} = e^x$$

$$v = \frac{e^x}{1+e^x} //$$

$$(ii) \quad \frac{dn}{dt} = \frac{e^{-n}}{1+e^n}$$

$$\int \frac{1+e^n}{e^n} dn = \int dt$$

$$-e^{-n} + n = t + C$$

$$t \rightarrow 0, n \rightarrow 0$$

$$\therefore C = -1$$

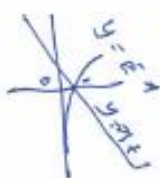
$$\Rightarrow t = 1 + n - e^{-n}$$

$$(iii) \quad \text{as } t \rightarrow 0$$

$$1 + n - e^{-n} \rightarrow 0$$

$$\Rightarrow n+1 > e^{-n}$$

$$\Rightarrow n > 0$$



$$\left. \begin{aligned} v &= \frac{e^x}{1+e^x} \\ &= \frac{1}{1+e^{-x}} \end{aligned} \right\} \begin{aligned} \text{as } x > 0 \\ \frac{1}{2} < \frac{1}{1+e^{-x}} < 1 \Rightarrow \frac{1}{2} < v < 1 \end{aligned}$$

$$(C) \quad (i) \quad x = v \cos \theta \cdot t \Rightarrow t = \frac{x}{v \cos \theta}$$

$$\text{Sub in } t = \frac{x}{v \cos \theta} \text{ into } y$$

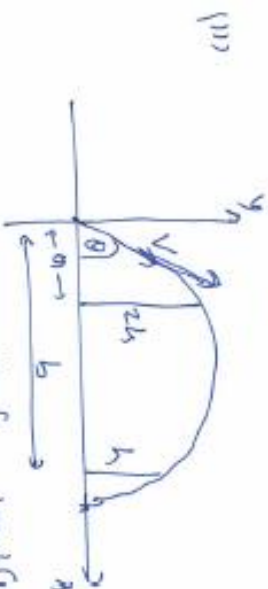
$$y = v \sin \theta \left(\frac{x}{v \cos \theta} \right) - \frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2$$

$$= x \tan \theta - \frac{g x^2}{2 v^2 \cos^2 \theta}$$

$$= x \tan \theta - \frac{g x^2 \sec^2 \theta}{2 v^2}$$

$$= x \tan \theta - \frac{g x^2 (1 + \tan^2 \theta)}{2 v^2}$$

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Applying the result from part (i),

when the particle clears the wall of height h , $2h$ when

$$2h = a \tan \theta - \frac{g x^2}{2 v^2} (1 + \tan^2 \theta)$$

$$\Rightarrow 2h - a \tan \theta = - \frac{g x^2}{2 v^2} (1 + \tan^2 \theta) \quad \text{--- (1)}$$

when the particle clears the wall of height h when

$$h = b \tan \theta - \frac{g b^2}{2 v^2} (1 + \tan^2 \theta) \quad \text{--- (2)}$$

$$\Rightarrow h - b \tan \theta = - \frac{g b^2}{2 v^2} (1 + \tan^2 \theta)$$

$$\text{①} \div \text{②} \quad \frac{2h - a \tan \theta}{h - b \tan \theta} = \frac{a^2}{b^2}$$

$$\Rightarrow \tan \theta = \frac{2hb^2 - ha^2}{ab^2 - a^2b} = \frac{h(2b^2 - a^2)}{ab(b - a)}$$

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